Hanik **Lakhe**

Water Engineering and Management Graduate Student ·

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"Make your life meaningful and work hard to achieve what you want in your life."

Summary

Ambitious and dedicated graduate student in Water Engineering and Management at the Asian Institute of Technology (AIT), Thailand, with over five years of hands-on research experience in hydrology, environmental monitoring, GIS-based spatial analysis, and citizen science. Skilled in hydrological modelling, remote sensing, and field-based data collection across diverse environments, including Himalayan river systems and urban water settings in Nepal and Thailand. Proven track record of contributing to peer-reviewed publications and presenting at international conferences. Experienced in leading multidisciplinary research projects from design to dissemination, with strong analytical, communication, and problem-solving skills. Passionate about addressing water and climate challenges through innovative, data-driven approaches, and committed to advancing sustainable water resource management within dynamic, impact-focused organizations.

Education

Asian Institute of Technology
Paholyothin Highway, Klong Luang, Pathum Thani 12120 Thailand

Master in Water Engineering and Management — *2024 - 2026*

- Got a 75 percent AIT Scholarship and 25 percent Global Water and Sanitation Center (GWSC) Scholarship + Two registration waivers
- Got 3.87 out of 4 GPA in the coursework (based on first and second semester

Khwopa College, Tribhuvan University
Bhaktapur, Nepal

B.Sc. in Environmental Science — *2017 - 2021*

- First Division
- Got an Academic Merit Scholarship due to previous Intermediate (+2) Level performance and highest marks in the entrance examination

Work Experience

Global Water and Sanitation Center (GWSC)
Pathum Thani, Thailand

Student Research Assistant — *June 2025 - present*

- Prepare and manage geospatial datasets related to sanitation infrastructure, service coverage, and urban demographics.
- Generate maps using ArcGIS or QGIS for CWIS city assessments, investment planning, and presentations.
- Support spatial analysis to identify service gaps and infrastructure risks.
- Integrate field data and geo-referenced surveys into project dashboards or planning tools.
- Assist in developing technical reports, knowledge products, and visual materials based on GIS outputs.
- Participate in event preparation, Center-wide workshops, and cross-team documentation efforts.
- Support other tasks assigned by the TA Hub or GWSC management team.

Asian Infrastructure Investment Bank (AIIB) — under Dr. Srinivasan Ancha
Remote

Freelance GIS & Climate Risk Consultant — *August 2025 – September 2025*

- Conducted a comprehensive climate risk assessment of road infrastructure across 10 provinces in Cambodia, supporting AIIB-funded climate resilience and infrastructure planning initiatives.
- Developed high-resolution multi-hazard risk maps for 38 priority road segments using advanced GIS and spatial analysis techniques.
- Assessed five major climate hazards—flood, drought, landslide, heatwave, and coastal flooding—to evaluate both current and future risk.
- Applied a top-down climate modelling framework using historical baseline data (1995–2014) and future climate projections under SSP2-4.5 and SSP5-8.5 scenarios for the 2040–2059 period.
- Integrated geospatial datasets, climate model outputs, and infrastructure data to analyze identify high-risk roads.
- Designed and implemented a multi-hazard risk classification framework combining multiple factors and projected climate impacts to quantify risk levels.
- Generated clear, decision-ready visualizations and maps to communicate complex climate risks to technical and non-technical stakeholders.
- Delivered geospatial datasets, analytical outputs, and technical documentation to support evidence-based investment planning and climate adaptation strategies.

Water Engineering and Management | School of Engineering and Technology | Asian Institute of Technology
Pathum Thani, Thailand

Research Assistant/ Modeler *Jan 2025 - Jun 2025*

- Develop water models (SWAT) for effective use in Northern Thailand.
- Create a water security framework and identify nature-based solutions.
- Conduct case studies on NBS and document best practices for urban water management.
- Organize project meetings and act as a point of contact among research partners.
- Support fieldwork, research, teaching, and workshop activities

Smartphones For Water Nepal
Thasikhel, Nepal

Research Associate | Research Assistant | Research Assistant Intern *Jul 2022 - Jul 2024 | Jan 2019 - Jun 2022 | Jun - Dec 2019*

- Managed overall S4W-Nepal data and ensured quality control, including regular data validation and updates.
- Facilitated a research group (Surface Water) and supported idea development, proposal writing, and implementation.
- Conducted field data collection, site visits, and performed spatial and statistical data analysis.
- Authored scientific articles, citizen science stories, presentations, and contributed to peer-reviewed publications.
- Represented S4W-Nepal in meetings, conferences, and programs as directed by the CEO.
- Recruited, trained, and supervised citizen scientists and volunteers, enhancing project reach and engagement.
- Oversaw organizational programs, awareness campaigns, and managed social media platforms (Facebook, Instagram, YouTube).
- Prepared monthly progress reports for activities, campaigns, and strategic projects, including actionable insights.
- Strengthened collaborations with stakeholders through effective communication and outreach initiatives.

Monitoring and Evaluating Suspended Sediment in Himalaya (MESSHI) Project

Project Leader *Jan 2023 - July 2024*

- Developed and implemented project plans, monitoring protocols, and laboratory procedures for suspended sediment analysis.
- Established a laboratory for suspended sediment determination and installed sensors at monitoring sites, ensuring real-time data collection.
- Led fieldwork planning, data collection efforts, and ensured meticulous data storage and management.
- Recruited, trained, and managed a team of citizen scientists and volunteers for field and lab activities.
- Coordinated with stakeholders, collaborators, and donors to align project goals and ensure seamless execution.
- Authored detailed project reports, scientific publications, and contributed to conference papers to disseminate findings.
- Prepared comprehensive progress reports and technical documentation for stakeholders, ensuring transparency and accountability.
- Facilitated workshops and training sessions for stakeholders, fostering knowledge sharing and capacity building.

HIMALAYAN CLIMATE DATA FIELD LAB 2024
Entrance Cafe, Patan

Participant *May 13th – June 7th, 2024*

- Collaborated with research groups, including Sensing and Sensors, Chitwan Flooding, AI-Machine Learning, and Re-imagining Resilience and Vulnerability.
- Conducted vulnerability assessments and developed post-disaster management plans for informal settlements along the Bagmati River Corridor using drone and 360-degree imagery.
- Developed a machine learning model to create a landslide susceptibility map, enhancing hazard risk analysis.
- Participated in interactive sessions on disaster risk management, resilience building, and climate adaptation.
- Attended workshops on GitHub, Google Earth Engine, and advanced disaster risk modeling techniques.
- Networked with research enthusiasts and interdisciplinary professionals from over 17 countries to foster collaborations.
- Prepared a comprehensive summary report highlighting key learnings and actionable recommendations for field applications.

Young Researchers' Circle
Thasikhel, Nepal

Treasurer | Coordinator *Apr 2020 - Aug 2021 | Aug 2021 - Jul 2022*

- Managed financial oversight, including budget estimation, fund mobilization, and financial reporting to ensure transparency.
- Tracked expenses, prepared financial summaries, and provided regular updates to the executive committee.
- Led a group of 50+ early career research enthusiasts, graduate, and undergraduate students, fostering collaborative learning.
- Coordinated and supervised the work of YRC members, ensuring alignment with organizational goals and objectives.
- Involved in strategic planning, project execution, and tracking progress of YRC initiatives and campaigns.
- Maintained updated records, prepared reports, and followed up on decisions from executive committee meetings.
- Contributed to the bimonthly "Young Researcher" newsletter and fortnightly "Environmental News Refresher" (ENR).
- Organized training programs and seminars to enhance research capacity and engagement among members.

Skills

Programming Skills	Intermediate Level Python, AI, and Machine Learning, Intermediate Level Google Earth Engine, Basic R Studio
Spatial Analysis Tools	Intermediate Level Quantum GIS and ArcGIS
Modelling Softwares	HEC-HMS, HEC-RAS, HEC-Res Sim, MIKE+ 2025, SWAT, RRI, Dyna-clue Landuse Modeling and Climate Change modeling
Water Accounting	Hydrology, field data collection, data management, benthic sampling, sorting, and identification
Environmental Monitoring	Acoustic Doppler Velocimeter, OBS sensors, Auto level, Data loggers and other equipment, Water Quality
Basic Softwares	Open Data Kit (ODK) Collect, ODK Aggregate, HOBOware, Microsoft Office Suite, and LaTeX
Languages	English, Nepali, Hindi, Newari

Publications

Published Paper

Prajapati, R., Karki, S., Thapa, S., **Lakhe, H.**, Bain, D., Gardner, J. (2026). Suspended sediment dynamics in an urban, mountain catchment in Nepal. EGUsphere, 2026, 1-35.

Duwal, S., Prajapati, R., Upadhyay, S., Neupane, S., **Lakhe, H.**, Thapa, B. R., Davids .J.C. Talchabhadel, R. (2025). Leveraging a Citizen Science Approach for Rainfall Monitoring: Evaluating Performance and Reliability To Complement Standard Datasets. Earth Systems and Environment, 1-16.

Upadhyay, S., Silwal, P., Prajapati, R., Talchabhadel, R., Shrestha, S., Duwal, S. and **Lakhe, H.** (2022) Evaluating Magnitude Agreement and Occurrence Consistency of CHIRPS Product with Ground-Based Observations over Medium-Sized River Basins in Nepal. Hydrology, 9(8), pp.146. Available at DOI: https://doi.org/10.3390/hydrology9080146

Kindermann, P.E., Brouwer, W.S., van Hamel, A., van Haren, M., Verboeket, R.P., Nane, G.F., **Lakhe, H.**, Prajapati, R. and Davids, J.C. (2020) Return level analysis of the Hanumante river using structured expert judgment: A reconstruction of historical water levels. Water, 12(11), pp. 3229. Available at DOI: https://doi.org/10.3390/w12113229

IN PROGRESS

Anticipating Extreme Temperature Indices in Nepal: Insights from the CMIP6 Climate Model Assessment.

Conference proceedings

Prajapati, R., Gardner, J., Lakhe, H. (2024, June). Linking satellite, sensor, and citizen science to understand sediment dynamics in Himalayan rivers. In Water Science Conference (WaterSciCon24 (No. 227, pp. 223-227).

Duwal, S., **Lakhe, H.**, Neupane, S., Prajapati, R. (2024, June). Evaluating Citizen Scientist's Performance on Water Resources Observation in the Kathmandu Valley. In Water Science Conference (WaterSciCon24 (No. 24, pp. 312-024).

Prajapati, R., Gardner, J. and **Lakhe, H.**, 2023. Quantifying Suspended Sediment Flux in Himalayan Rivers of Nepal using Citizen Science and Low-Cost Sensors. AGU23.

Mandal, R.B., Shakya, S., Ghale, B., **Lakhe, H.**, Duwal, S. and Shakya, B., 2023. Assessing Groundwater Level and Water Quality in Bhaktapur Municipality: A Pre-Monsoon and Post-Monsoon Analysis. Khwopa Journal, 5(2), pp.74-90.

Shakya S., Ghale B., **Lakhe,H.,** Duwal S., Chalise A., Mandal B.R., and Shakya, M.B., (2022). Hiti System in Bhaktapur: An Evaluation of its Current Status and Importance KEC Conference 2023.

Lakhe, H., Silwal, P., Chalise, A., Duwal, S., Shakya, S., Prajapati, R., and Shakya, B. M. (2022, December) Application of Water Quality Index for Groundwater Quality Assessment and its Perspective among the Urban Dwellers of the Kathmandu Valley, Nepal. In Fall Meeting 2022. AGU.

Mandal, R. B., **Lakhe, H.**, Duwal, S., Chalise, A., Shakya, S., Prajapati, R. and Shakya, B. M. (2022, December) Evaluation of Interconnection between Streamflow and Rainfall in Headwater Streams of the Kathmandu Valley. In Fall Meeting 2022. AGU.

Duwal, S., Shakya, B.M, Chalise, A., **Lakhe, H.**, Prajapati, R. and Talchabhadel, R. (2022, December) Evaluation of Citizen Science-Based Rainfall Data with Standard Ground-Based Rain Gauge and Remotely Sensed Rainfall Data in Kathmandu Valley. In Fall Meeting 2022. AGU.

Shakya, S., **Lakhe, H.**, Chalise, A., Duwal, S., Silwal, P., Prajapati, R., Shakya, B. M. (2022, December) Assessment of the Seasonal Interaction between Streams and Aquifers of the Kathmandu Valley. In Fall Meeting 2022. AGU.

Ghale, B., **Lakhe, H.**, Duwal, S., and Shakya, B. M. (2022, December) Convective clouds and associated severe weather events over Kathmandu Valley, Nepal. In Fall Meeting 2022. AGU.

Lakhe,H., Silwal, P., Duwal, S., Chalise, A., Shakya S. and Shakya, M.B., (2022). Application of the Water Quality Index (WQI) for Irrigational Water Quality Assessment of the Hanumante River. KEC Conference 2022.

Chalise A., Silwal, P., **Lakhe,H.**, Duwal, S., Shakya S. and Shakya, M.B., (2022). Evaluating the Influence of Rainfall and Land Use Type on the Shallow Groundwater Level Fluctuations of Bhaktapur District. KEC Conference 2022.

Lakhe,H., Silwal, P., Duwal, S., Prajapati, R., Shakya, S., and Davids, J.C., (2021), December. Estimation of the Total Baseflow of the Kathmandu Valley Using Monthly Streamflow Measurements from a Sub-Sample of Headwater Catchments. In AGU Fall Meeting 2021. AGU.

Silwal, P., Upadhyay, S., Duwal, S., **Lakhe,H.**, Prajapati, R. and Talchabhadel, R., (2021), December. Evaluation and Comparison of Satellite Based-Rainfall Product and Ground Rain Gauge Observations over Five Basins of Nepal. In AGU Fall Meeting 2021. AGU.

Duwal, S., Chalise, A., Prajapati, R., Silwal, P., **Lakhe,H.**, Talchabhadel, R., (2021), December. Citizen Scientists Measure Rainfall of Kathmandu Valley Comparable to Ground-Based Rain Gauge Observations and Satellite Rainfall Estimates. In AGU Fall Meeting 2021. AGU.

Lakhe,H., Silwal, P., Duwal, S., Chalise, A. and Prajapati, R., (2020). Assessing the linkage between streamflow and rainfall in the headwater streams of the Kathmandu Valley. KEC Conference 2020.

Duwal, S., Silwal, P., **Lakhe,H.**, Chalise, A. and Prajapati, R., (2020). Reliability of Citizen Science-based Rainfall Monitoring in the Kathmandu Valley. KEC Conference 2020.

Lakhe,H., Upadhyay, S., Thapa, A.B., Prajapati, R. and Davids, J.C., (2019). Observing our Headwaters: Lessons from 2 Years of Community Monitoring of the Upper Bagmati and Nagmati Watersheds, 4th International Conference on Mountains in the Changing World, Kathmandu, Nepal, 18-19 October 2019

Academic Projects & Coursework

Master of Science in Water Engineering and Management

Asian Institute of Technology (AIT), Thailand (2024–present)

Master's Thesis

Towards Future Flood Hazard Mapping under Climate and Land Use Change Using Integrated LSTM and HEC-RAS Modeling: Karnali River Basin, Nepal.

Academic Advisor: Prof. Dr. Sangam Shrestha

Tools: LSTM (Deep Learning), HEC-RAS, GIS, Climate & Land Use Change Modelling

Status: Ongoing

Term Reports

1. Landslide Susceptibility Mapping in Nepal Using Machine Learning and NASA's Global Landslide Catalog.

2. A Novel Metric-Based Assessment of Drought Stress: A Case Study in Chiang Rai Province, Thailand (2000–2021). Developed a composite drought stress metric using multi-source remote sensing and ground-based data over a 21-year period.

Course Projects

1. Assessing the Risk of Failure of Water Supply System during the Dry Season in Kelani River Basin, Sri Lanka. Evaluated dry-season supply reliability and identified critical failure points in the Kelani River water supply network.

2. Reservoir System Operation Studies for Development of Proposed Power Plant in Saraburi Province, Pasak River Basin, Thailand. Simulated reservoir operation scenarios to optimize water release strategies for hydropower generation using HEC-ResSim.

3. Hydrodynamic Modelling of Lower Chao Phraya River Basin, Thailand using MIKE+ River. Built and calibrated a 1D hydrodynamic model of the Lower Chao Phraya to simulate flood inundation extents and water levels.

4. Stream Water Quality Modelling using Qual-2K Model. Applied the Qual-2K model to simulate dissolved oxygen, BOD, and nutrient dynamics along a stream reach under varying pollution loads.

5. Groundwater Flow Modelling using Visual MODFLOW. Constructed a 3D groundwater flow model using Visual MODFLOW to simulate aquifer behaviour and assess recharge and discharge zones.

Relevant Coursework

Watershed Hydrology, Hydrodynamics, Water Resources Systems, Concepts in Water Modelling, AI & Big Data in Water, Research Design and Experimental Methods, Groundwater Development and Management, Modelling of Water Resources Systems, Earth Engine for Water Resources Monitoring and Management, Hydroinformatics and Modern Hydrological Modelling.

Bachelor of Science in Environmental Science

Khwopa College, Tribhuvan University, Nepal (2017–2021)

Bachelor's Thesis

Assessment of Water Quantity and Quality of Traditional Stone Spouts ('Hitis') of the Bhaktapur Municipality.

Academic Advisor: Mrs. Meera Prajapati

Tools: GIS and Water Quality Index (WQI)

Term Reports

1. Initial Environmental Examination (IEE) of Upper Khimti Hydropower Project (7 MW), Dolakha and Ramechhap, Bagmati Province, Nepal. Conducted a systematic IEE covering baseline environmental conditions, impact prediction, and mitigation measures for a proposed run-of-river hydropower scheme in the Bagmati Province.

Relevant Coursework

Introduction to Environmental Science, Fundamental Environmental Science, Climate Change, Environmental Pollution & Management, Solid Waste Management, Res. & Biodiversity Conservation, Pollution Monitoring & Control, Scientific Communication, Research Methodology and Applied Statistics

Selected Award

AGU Fall Meeting Travel Grant — *American Geophysical Union, Washington, USA*

Grantee — *2021*

MoChWo Scholarship — *Kathmandu Institute of Applied Sciences, Kathmandu, Nepal*

Scholarship holder — *2019*

Mahatma Gandhi Scholarship — *Indian Embassy, Kathmandu, Nepal*

Scholoarship holder — *2015*

References

Prof. Dr. Sangam Shrestha

Professor, Water Engineering and Management, Faculty of Civil and Environmental Engineering, Asian Institute of Technology (AIT)

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Contact No: +66 2 524 6055

Dr. Rocky Talchabhadel

Assistant Professor, Jackson State University

Vice President at Smartphones For Water Nepal (S4W-Nepal)

Email: rocky.talchabhadel@jsums.edu, rocky.ioe@gmail.com

Dr. Srinivasan Ancha

Advisor (Agriculture, Food, Nature, and Rural Development), Asian Development Bank (ADB)

Senior Fellow, Institute for Global Environmental Strategies (IGES)

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Dr. Sanjiv Neupane

CEO at Smartphones For Water Nepal (S4W-Nepal) (Formerly)

Data Innovation Hub Manager, Global Water and Sanitation Center (GWSC)

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